HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	Quiz 5 SOLUTION	Marks 30 Weight 4%	April 20, 2026

Important Notes

Clearly write your name and student ID on top.

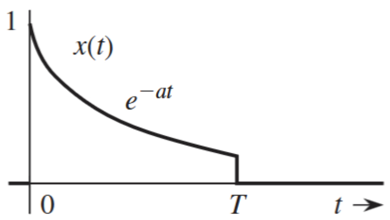
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)

Operation	$x(t)$	$X(\omega)$
Scalar multiplication	$kx(t)$	$kX(\omega)$
Addition	$x_1(t)+x_2(t)$	$X_1(\omega)+X_2(\omega)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Scaling (a real)	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{\omega}{a}\right)$
Time shifting	$x(t-t_0)$	$X(\omega)e^{-j\omega t_0}$
Frequency shifting (ω_0 real)	$x(t)e^{j\omega_0 t}$	$X(\omega-\omega_0)$
Time convolution	$x_1(t)*x_2(t)$	$X_1(\omega)X_2(\omega)$

$$X(\omega)=\int_{-\infty}^{\infty}x(t)e^{-j\omega t}dt$$

Problem 1 [10 pts]

Find the Fourier Transform of the signal shown



Solution:

Using

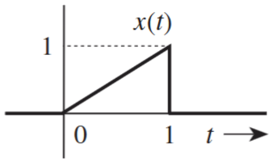
$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

We get

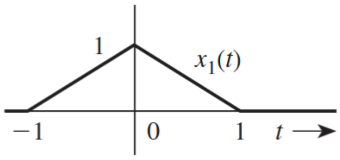
$$X(\omega) = \int_0^T e^{-at}e^{-j\omega t}\,dt = \int_0^T e^{-(j\omega+a)t}\,dt = \frac{1-e^{-(j\omega+a)T}}{j\omega+a}$$

Problem 2 [10 pts]

Given that $x(t)$ has Fourier Transform $X(\omega)$, find the Fourier Transform of $x_1(t)$.



$$X(\omega) = \frac{1}{\omega^2}(e^{j\omega} - j\omega e^{j\omega} - 1)$$



Solution:

We first note that

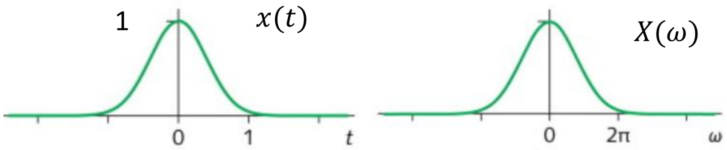
$$x_1(t) = x(t+1) + x(-t+1).$$

Using the time-shift and time-reflection properties yields

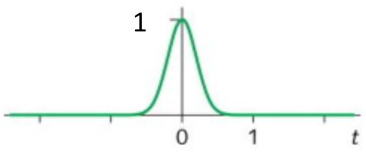
$$X_1(\omega) = X(\omega)e^{j\omega} + X(-\omega)e^{-j\omega}.$$

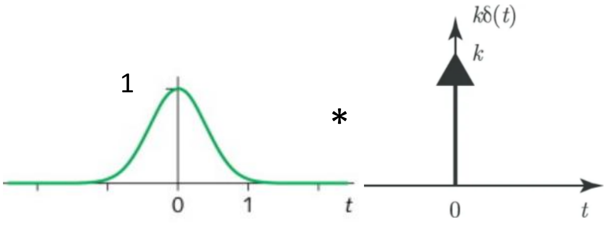
Problem 3 [10 pts]

Consider $x(t)$ and its Fourier Transform $X(\omega)$ below



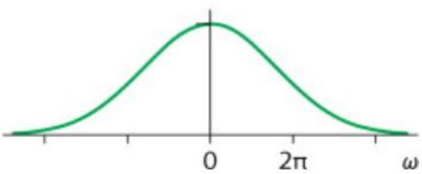
Draw rough sketches of the Fourier Transforms of the following (justify your sketch in each case)





Solution:

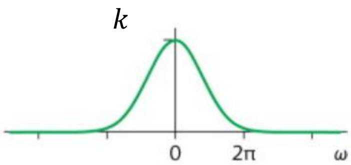
(a) Since damping causes spread, we have



(b) We first note that delta is infinitely damped and becomes straight line at height k in frequency.

Next, we use the property that convolution in time becomes product in frequency.

Product becomes $kX(\omega)$.



Another way to see it is to recall that impulse is the identity of convolution.